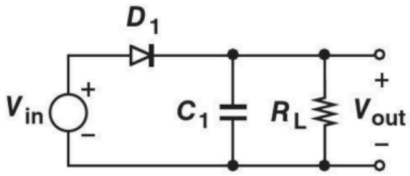
Midterm

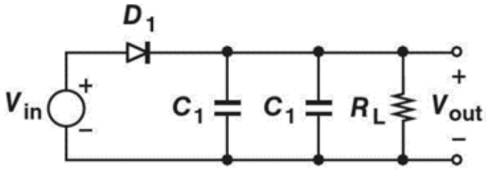
5 Problems (20 points each, total 100 points)

學號 515661055　陳欣怡

1. The half-wave rectifier shown below drives a 100-Ω load through a diode with $V_{D,on} = 0.6\ \text{V}$ and a 500-μF smoothing capacitor as shown in (a). The input is a 60-Hz sinusoidal voltage source with a peak voltage of 3.5 V. If the circuit is replaced by (b), Calculate the ripple amplitude respectively (20 points)



(a)



(b)

$V_{out,\text{peak}} = V_p - V_{D,on} = 3.5 - 0.6 = 2.9\ \text{V}$

$I_L \approx \dfrac{V_{out,\text{peak}}}{R_L} = \dfrac{2.9\ \text{V}}{100\ \Omega} = 29\ \text{mA}$

$V_{r,pp} \approx \dfrac{I_L}{C\, f_{in}}$　　(小漣波近似，$V_{r,pp}$ 為峰對峰值)

(a)　$V_{r,pp} \approx \dfrac{29 \times 10^{-3}}{(500 \times 10^{-6})(60)} = 0.967\ \text{V}$

(b)　$C_{eq} = C_1 + C_1 = 1000\ \mu\text{F} \ \Rightarrow\ V_{r,pp} \approx \dfrac{29 \times 10^{-3}}{(1000 \times 10^{-6})(60)} = 0.483\ \text{V}$

Ans. (a) $V_{r,pp} \approx 0.967\ \text{V}$　　(b) $V_{r,pp} \approx 0.483\ \text{V}$

2. A junction employs $N_D = 4 \times 10^{16}\ \text{cm}^{-3}$ and $N_A = 3 \times 10^{15}\text{cm}^{-3}$ Calculate the built-in potential at (a) T = 300 K and (b) 350 K. (20 points)

$$V_T = \frac{kT}{q}, \qquad n_i = 5.2 \times 10^{15}\, T^{3/2} \exp\left[-\frac{E_g}{2kT}\right], \qquad V_0 = V_T \ln \frac{N_A N_D}{n_i^2}$$

Si : $E_g = 1.12\ \text{eV}, \quad k = 8.625 \times 10^{-5}\ \text{eV/K}, \quad k/q = 8.625 \times 10^{-5}\ \text{V/K}$

(a) $T = 300\ \text{K}$: $\quad V_T = 0.025875\ \text{V}, \quad n_i = 1.0776 \times 10^{10}\ \text{cm}^{-3}$

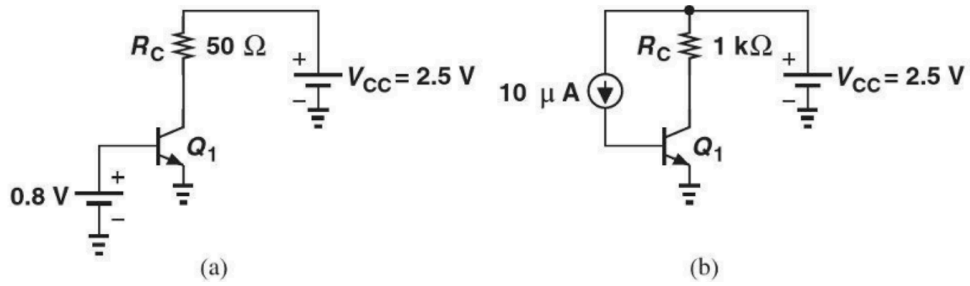
$$V_0 = (0.025875) \ln \frac{(3 \times 10^{15})(4 \times 10^{16})}{(1.0776 \times 10^{10})^2} = 0.715802\ \text{V} \approx 0.716\ \text{V}$$

(b) $T = 350\ \text{K}$: $\quad V_T = 0.0301875\ \text{V}, \quad n_i = 2.9897 \times 10^{11}\ \text{cm}^{-3}$

$$V_0 = (0.0301875) \ln \frac{(3 \times 10^{15})(4 \times 10^{16})}{(2.9897 \times 10^{11})^2} = 0.634476\ \text{V} \approx 0.634\ \text{V}$$

Ans. (a) $V_0 \approx 0.716\ \text{V}$ (b) $V_0 \approx 0.634\ \text{V}$

3. Determine the operating point and the small-signal model of Q for the circuits shown in (a) and (b). Assume $I_S = 2\text{x}10^{-16}$ A, $\beta = 100$, and $V_A = \infty$. (20points)



(a) $V_{BE} = 0.8\ \text{V}$

$I_C = I_S e^{V_{BE}/V_T} = (2 \times 10^{-16}) e^{0.8/0.026} = 4.61\ \text{mA}$

$I_B = \dfrac{I_C}{\beta} = 46.1\ \mu\text{A}, \qquad I_E = I_C + I_B = 4.66\ \text{mA}$

$V_C = V_{CC} - I_C R_C = 2.5 - (4.61\ \text{mA})(50\ \Omega) \approx 2.269\ \text{V}$

$V_C \approx 2.269\ \text{V} > V_B = 0.800\ \text{V} > V_E = 0 \ \Rightarrow$ 順向主動區

$g_m = \dfrac{I_C}{V_T} \approx 177\ \text{mS}, \qquad r_\pi = \dfrac{\beta}{g_m} \approx 564\ \Omega, \qquad r_o = \dfrac{V_A}{I_C} = \infty$

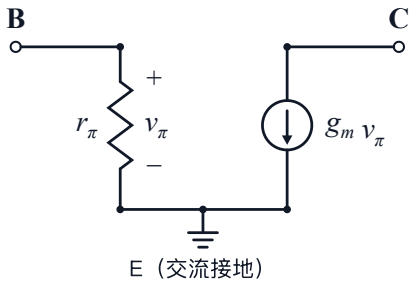
(b) $I_B = 10\ \mu\text{A}$

$I_C = \beta I_B = 1\ \text{mA}, \qquad I_E = 1.01\ \text{mA}, \qquad V_C = 2.5 - (1\ \text{mA})(1\ \text{k}\Omega) = 1.500\ \text{V}$

$V_{BE} = V_T \ln \dfrac{I_C}{I_S} = (26\ \text{mV}) \ln \dfrac{10^{-3}}{2 \times 10^{-16}} \approx 760\ \text{mV}$

$V_C = 1.500\ \text{V} > V_B \approx 0.760\ \text{V} > V_E = 0 \ \Rightarrow$ 順向主動區

$g_m = \dfrac{I_C}{V_T} \approx 38.5\ \text{mS}, \qquad r_\pi = \dfrac{\beta}{g_m} = 2.6\ \text{k}\Omega, \qquad r_o = \infty$



(a) $g_m \approx 177\ \text{mS}, \quad r_\pi \approx 564\ \Omega, \quad r_o = \infty$

(b) $g_m \approx 38.5\ \text{mS}, \quad r_\pi = 2.6\ \text{k}\Omega, \quad r_o = \infty$

$r_o = V_A / I_C = \infty$（開路，忽略）

Ans. (a) $I_C = 4.61\ \text{mA},\ I_B = 46.1\ \mu\text{A},\ V_{CE} \approx 2.269\ \text{V};\ \ g_m \approx 177\ \text{mS},\ r_\pi \approx 564\ \Omega$
(b) $I_C = 1\ \text{mA},\ V_{BE} \approx 760\ \text{mV},\ V_{CE} = 1.500\ \text{V};\ \ g_m \approx 38.5\ \text{mS},\ r_\pi = 2.6\ \text{k}\Omega$

4. A bipolar transistor carries a collector current of 1mA with $V_{CE} = 2V$. Determine the required base-emitter voltage if (a) $V_A = \infty$ or (b) $V_A = 20V$. Assume $I_S = 3 \times 10^{-16}$ A.

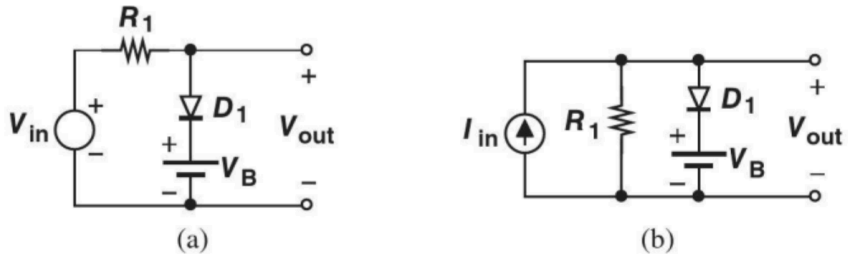
$$I_C = I_S e^{V_{BE}/V_T}\left(1 + \frac{V_{CE}}{V_A}\right) \Rightarrow V_{BE} = V_T \ln\left(\frac{I_C}{I_S} \cdot \frac{1}{1 + V_{CE}/V_A}\right)$$

(a) $V_A = \infty$: $\quad V_{BE} = (26\ \text{mV}) \ln \dfrac{10^{-3}}{3 \times 10^{-16}} = 750\ \text{mV}$

(b) $V_A = 20\ \text{V}$: $\quad 1 + \dfrac{2}{20} = 1.1, \qquad V_{BE} = (26\ \text{mV}) \ln\left(\dfrac{10^{-3}}{3 \times 10^{-16}} \cdot \dfrac{1}{1.1}\right) = 747\ \text{mV}$

Ans. (a) $V_{BE} = 750\ \text{mV}$ $\qquad$ (b) $V_{BE} = 747\ \text{mV}$

5. In the circuits of (a) and (b), plot the current flowing through R_1 as a function of V_{in} *or* I_{in}. Assume a constant-voltage diode model with $V_{Don} = 0.8V$ and $V_B = 1V$ (20points)

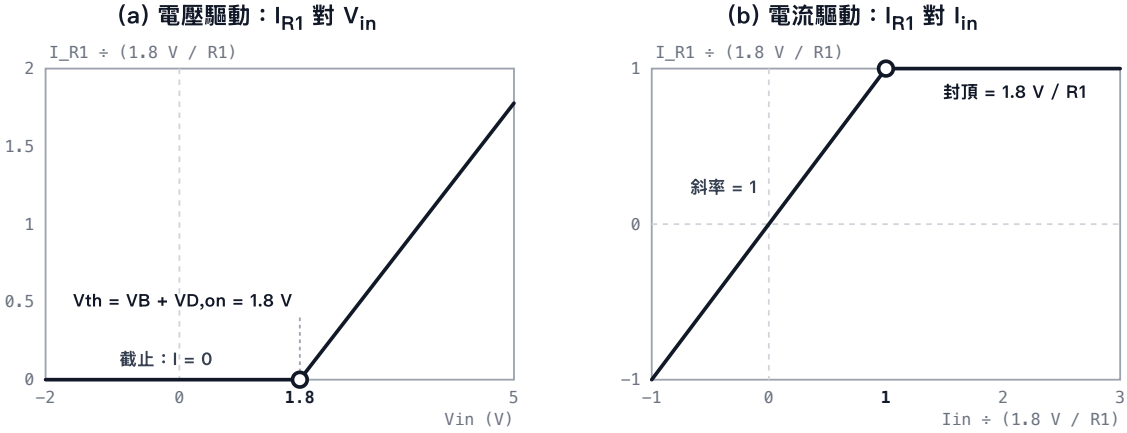


(a) (b)

D_1 on : $V_{out} \geq V_B + V_{D,on} = 1\ \text{V} + 0.8\ \text{V} = 1.8\ \text{V}$

(a) $V_{in} < 1.8\ \text{V}:\ I_{R_1} = 0; \qquad V_{in} \geq 1.8\ \text{V}:\ I_{R_1} = \dfrac{V_{in} - 1.8\ \text{V}}{R_1}$

(b) $I_{in} < 1.8\ \text{V}/R_1:\ I_{R_1} = I_{in}; \qquad I_{in} \geq 1.8\ \text{V}/R_1:\ I_{R_1} = 1.8\ \text{V}/R_1$



(a) 圖縱軸已用 $1.8\ \text{V}/R_1$ 正規化，實際斜率 $\text{d}I_{R_1}/\text{d}V_{in} = 1/R_1$。

Ans. (a) 轉折於 $V_{in} = 1.8\ \text{V}$，之後 $I_{R_1} = (V_{in} - 1.8\ \text{V})/R_1$
(b) 轉折於 $I_{in} = 1.8\ \text{V}/R_1$，之後定值 $1.8\ \text{V}/R_1$